Title: BMW Global Sales Performance & EV Transition Analysis (2018-2025)

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Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn), Tableau.

1. Executive Summary This case study explores BMW's global sales trajectory from 2018 to 2025, evaluating the performance of internal combustion engine (ICE) models alongside the growing Battery Electric Vehicle (BEV) lineup (i4, iX). The analysis extracts actionable insights regarding revenue generation, premium market share, and regional economic factors influencing sales.

2. Business Problem & Objectives The automotive industry is undergoing a massive shift towards electrification. The objective of this project was to:

- Analyze sales volume and revenue trends across different BMW models (3 Series, 5 Series, X3, X5, X7, i4, iX).
- Track the growth of BEV market share versus traditional ICE models.
- Evaluate the impact of macroeconomic factors (GDP Growth, Fuel Price Index) on premium vehicle sales.
- Build a comprehensive, interactive dashboard for stakeholders to monitor KPIs.

3. Methodology & Data Processing The raw dataset (`bmw_globalsales_2018_2025.xlsx`) required rigorous preparation before visualization:

- **Data Cleaning (Python):** Utilized Pandas in a Jupyter/Colab environment to handle missing values, format data types, and aggregate monthly data into yearly trends.
- **Exploratory Data Analysis (EDA):** Leveraged Seaborn and Matplotlib to identify initial correlations between fuel prices and EV adoption rates.
- **Dashboarding (Tableau):** Engineered a dynamic Tableau workbook (`bmwt.twbx`) featuring KPI containers for Total Revenue, Average Price, and Units Sold, ensuring a highly structured and professional layout.

4. Key Insights & Findings

- **The Rise of the 'i' Series:** The data indicates a significant upward trend in the BEV Share (e.g., reaching up to 22.3% in specific regions by 2025), driven strongly by models like the i4 and iX.
- **SUV Dominance:** The X-series (X5, X7) consistently drives the highest revenue segments due to their higher average unit prices (Avg_Price_EUR) compared to the standard sedan models.
- **Economic Resilience:** Despite fluctuations in global GDP growth, BMW's premium share remained relatively stable, highlighting strong brand equity and targeted regional strategies.

5. Conclusion The analysis confirms BMW's successful strategic pivot towards electrification without cannibalizing its highly profitable luxury SUV segment. The accompanying Tableau dashboard provides an automated, real-time view of these dynamics, allowing decision-makers to track success rates and market penetration efficiently.